

Day -2 AIDS DAY-RUNS

Array

1] Array Basics

DS $\rightarrow$ arrays

$\rightarrow$ Def: Values

int defaults $\leftarrow [0, 0, 0, 0, 0]$

String defaults $\leftarrow [\text{null}, \text{null}, \text{null}]$

length is fixed: 5

$\rightarrow$ Random access

$a[2] = 30$ T.C $\rightarrow O(1)$

$\rightarrow$ Insert in middle T.C $\rightarrow O(n)$

insert 15 at 1: $[10, 15, 20, 30, 40, 0, 0, 0]$

shift = 3

insert 50 at end: $[10, 15, 20, 30, 40, 50, 0, 0]$

shift = 0

delete index 0: $[15, 20, 30, 40, 50, 0, 0, 0]$

shift = 1

$\rightarrow$ Bound check T.C $\rightarrow O(n)$

Bounds check: Index 8 out of bounds

→ Jagged array

T.C → $O(n^2)$

Jagged 2-D array (Pascal's triangle)

$x \rightarrow 0$ [1]

$x \rightarrow 1$ [1, 1]

$x \rightarrow 2$ [1, 2, 1]

$x \rightarrow 3$ [1, 3, 3, 1]

→ Array Copy

Arrays copy to grow: [15, 20, 30, 40, 50, 0, 0, 0, 0, 0, 0, 0, 0]

2) Array Stack

Array Stack of size 4

push 10 [10] ← top

" 20 [10, 20] "

" 30 [10, 20, 30] "

peek → 30

pop → 30 [10, 20] ← top

push 40, 50 [10, 20, 40, 50] ← top full? true

Stack overflow: capacity 4

50 40 20 10 ← popped in LIFO order

Stack underflow

3) Balanced Brackets - java

String [0] → balanced

" [1] → balanced

[2] → Not balanced

[3] → "

[4] → "

[5] → balanced

4) Big O Recap - java

q1 → $O(n)$

q1 → $O(n^2)$

q2 → $O(n)$

$O(n\sqrt{n})$

q3 → $O(\log n)$

q8 → (n^{26}) $O(n \log n)$

q4 → $O(n^2)$

1 1 2 3 4 5 6 6 n

q5 → $O(n \log n)$

2 2 3 6 3 n^2

q6 → $O(n)$

3 3 6 2 n^3

4 4 9 1 n^4

5 5 16 1 n^5

6 6 25 1 n^6

$\log_2 2 = 2$

$n(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}) \approx n \log n$

5] Circular Deque

addlast(B) Deque $\rightarrow$ [B] Row $\rightarrow$ [B, -, -] Front=0 Size=1
 " (C) " [B C] " [B, C, -] " Size=2
 addFirst(A) " $\rightarrow$ [A B C] " [B, C, - A] Front=3 Size=3
 addFirst(2) " $\rightarrow$ [2 A B C] " [B, C, 2, A] " = 2 " = 4
 addlast(D) capacity $\rightarrow$ 8 $\rightarrow$ [2, A B C D] " = 0 " = 5
 removeFirst() 2 x [A B C D]
 " last() D x [A B C]

6.] Circular Queue

enqueue (10) $\rightarrow$ (50) Size = 1, Size = 5 (Full)

dequeue x 3 10x, 20x, 30x size = 2

wrap $\rightarrow$ enqueue (60) at index 0 $\rightarrow$ [60, -, -, 40, 50]
 (70) [60, 70, -, 40, 50]
 (80) size full [60, 70, 80, 40, 50]
 90 x & Overflow

7.] Dynamic Array

Add 10, 20, 30, 40 Size = 4 Capacity = 4
 Add 50 Copy $\rightarrow$ 4 elem S $\rightarrow$ 6
 Add 60 S=6, C $\rightarrow$ 6
 Add 70 Copy $\rightarrow$ 6 elem 6 $\rightarrow$ 9
 90 S=9, C $\rightarrow$ 9
 100 Copy $\rightarrow$ 9 elem 9 $\rightarrow$ 13

Front Insert Race

Arraylist add(0, x) $O(n)$

LinkedList addFirst() $O(1)$

ArrayDeque addFirst() $O(1)$

Linear Queue

Linear Queue

Enqueue 10 $[10, _ _]$ $F=0, R=0$

" 20 $[10, 20, _]$ $R=1$

" 30 $[10, 20, 30, _]$ $R=2$

" 40 $[10, 20, 30, 40, _]$ $R=3$

" 50 $[10, 20, 30, 40, 50]$ $R=4$

dequeue 10 $[0, 20, 30, 40, 50]$ $F=1$

" 20 $[0, 0, 20, 40, 50]$ $F=2$

" 30 $[0, 0, 0, 40, 50]$ $F=3$

Postfix Evaluation

$3 + 4 * 2$

3	[]	3
+	[+]	3
4	[+]	3 4
*	[+*] [$*, +$]	3 4
2	[+*] [$*, +$]	3 4 2
End	E7	3 4 2 * +

Singly Linked List

head $\rightarrow$ null

tail $\rightarrow$ null

addlast(20)

h $\rightarrow$ 20 $\rightarrow$ null, tail $\rightarrow$ 20

addlast(30)

h $\rightarrow$ 20 $\rightarrow$ 30 $\rightarrow$ null, tail $\rightarrow$ 30

addFirst(10)

h $\rightarrow$ 10 $\rightarrow$ 20 $\rightarrow$ 30 $\rightarrow$ null, tail $\rightarrow$ 30

add(2, 25)

10 $\rightarrow$ 20 $\rightarrow$ 25 $\rightarrow$ 30

addlast(40)

10 $\rightarrow$ 20 $\rightarrow$ 30 $\rightarrow$ 40, tail $\rightarrow$ 40

remove(25)

10 $\rightarrow$ 20 $\rightarrow$ 30 $\rightarrow$ 40

removeFirst()

20 $\rightarrow$ 30 $\rightarrow$ 40

removeLast()

20 $\rightarrow$ 30

addlast(50), 60

20 $\rightarrow$ 30 $\rightarrow$ 50 $\rightarrow$ 60

Temp. java

$i \rightarrow 0$ elem $\rightarrow$ 10, 20, 30 llun $\leftarrow$ base
 $i \rightarrow 1$ 40, 50 Excepⁿ llun $\leftarrow$ list

a2.l $\rightarrow$ 3 (06) 40, 50, 60
but 2nd no $\rightarrow$ list col llun $\leftarrow$ 05 $\leftarrow$ d
(05) 40, 50, 60

05 $\leftarrow$ list llun $\leftarrow$ 05 $\leftarrow$ 05 $\leftarrow$ d
(01) 40, 50, 60

05 $\leftarrow$ list llun $\leftarrow$ 05 $\leftarrow$ 05 $\leftarrow$ d
(25, 5) /

05 $\leftarrow$ 75 $\leftarrow$ 05 $\leftarrow$ d
(0M) 40, 50, 60

0M $\leftarrow$ list 0M $\leftarrow$ 05 $\leftarrow$ 05 $\leftarrow$ d
(2M) 40, 50, 60

2M $\leftarrow$ 05 $\leftarrow$ 05 $\leftarrow$ d
(14, 9)